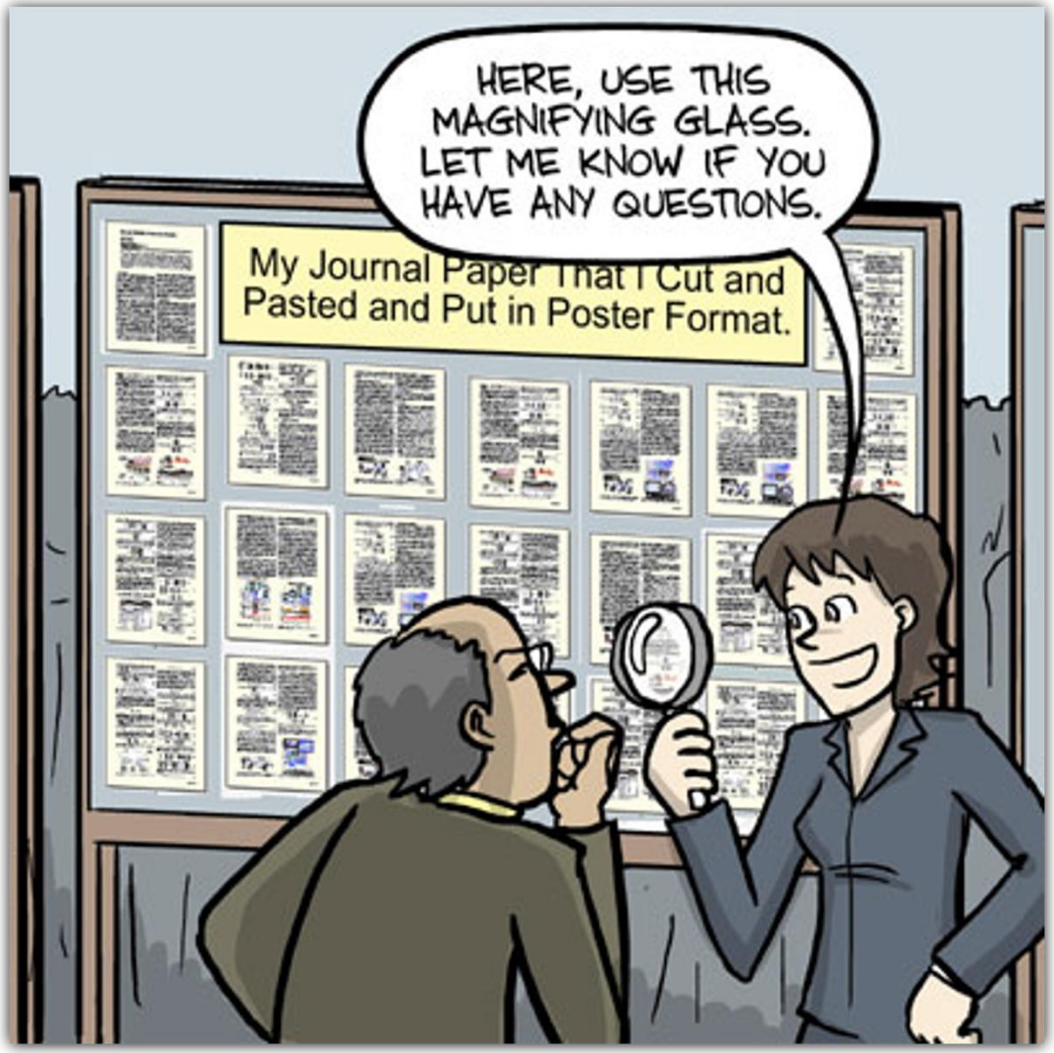
Posters

How Scientists Communicate

- People are tired
- Don’t put things on your poster that people will ignore
  - Note: People will ignore most things


HERE, USE THIS MAGNIFYING GLASS. LET ME KNOW IF YOU HAVE ANY QUESTIONS.
My Journal Paper That I Cut and Pasted and Put in Poster Format.

PIGS IN SPACE: EFFECT OF ZERO GRAVITY AND AD LIBITUM FEEDING ON WEIGHT GAIN IN CAVIA PORCELLUS

Colin B. Purrington*
6673 College Avenue, Swarthmore, PA 19081 USA

ABSTRACT:

One ignored benefit of space travel is a potential elimination of obesity, a chronic problem for a growing majority in many parts of the world. In theory, when an individual is in a condition of zero gravity, weight is eliminated. Indeed, in space one could conceivably follow ad libitum feeding and never even gain an gram, and the only side effect would be the need to upgrade one's stretchy pants("exercise pants"). But because many diet schemes start as very good theories only to be found to be rather harmful, we tested our predictions with a long-term experiment in a colony of Guinea pigs (Cavia porcellus) maintained on the International Space Station. Individuals were housed separately and given unlimited amounts of high-calorie food pellets. Fresh fruits and vegetables were not available in space so were not offered. Every 30 days, each Guinea pig was weighed. After 5 years, we found that individuals, on average, weighed nothing. In addition to weighing nothing, no weight appeared to be gained over the duration of the protocol. If space continues to be gravity-free, and we believe that assumption is sound, we believe that sending the overweight — and those at risk for overweight — to space would be a lasting cure.

INTRODUCTION:

The current obesity epidemic started in the early 1960s with the invention and proliferation of elastane and related stretchy fibers, which released wearers from the rigid constraints of clothes and permitted monthly weight gain without the need to buy new outfits. Indeed, exercise today for hundreds of million people involve only the act of wearing stretchy pants in public, presumably because the constrictive pressure forces fat molecules to adopt a more compact tertiary structure (Xavier 1965).

Luckily, at the same time that fabrics became stretchy, the race to the moon between the United States and Russia yielded a useful fact: gravity in outer space is minimal to nonexistent. When gravity is zero, objects cease to have weight. Indeed, early astronauts and cosmonauts had to secure themselves to their ships with seat belts and sticky boots. The potential application to weight loss was noted immediately, but at the time travel to space was prohibitively expensive and thus the issue was not seriously pursued. Now, however, multiple companies are developing cheap extra-orbital travel options for normal consumers, and potential travelers are also creating news ways to pay for products and services that they cannot actually afford. Together, these factors open the possibility that moving to space could cure overweight syndrome quickly and permanently for a large number of humans.

We studied this potential by following weight gain in Guinea pigs, known on Earth as fond of ad libitum feeding. Guinea pigs were long envisioned to be the "Guinea pigs" of space research, too, so they seemed like the obvious choice. Studies on humans are of course desirable, but we feel this current study will be critical in acquiring the attention of granting agencies.

MATERIALS AND METHODS:

One hundred male and one hundred female Guinea pigs (Cavia porcellus) were transported to the International Space Laboratory in 2010. Each pig was housed separately and deprived of exercise wheels and fresh fruits and vegetables for 48 months. Each month, pigs were individually weighed by duct-taping them to an electronic balance sensitive to 0.0001 grams. Back on Earth, an identical cohort was similarly maintained and weighed. Data was analyzed by statistics.

RESULTS:

Mean weight of pigs in space was 0.0000 +/- 0.0002 g. Some individuals weighed less than zero, some more, but these variations were due to reaction to the duct tape, we believe, which caused them to be alarmed push briefly against the force plate in the balance. Individuals on the Earth, the control cohort, gained about 240 g/month ($p = 0.0002$). Males and females gained a similar amount of weight on Earth (no main of effect of sex), and size at any point during the study was related to starting size (which was used as a covariate in the ANCOVA). Both Earth and space pigs developed substantial dewlaps (double chins) and were lethargic at the conclusion of the study.

CONCLUSIONS:

Our view that weight and weight gain would be zero in space was confirmed. Although we have not replicated this experiment on larger animals or primates, we are confident that our result would be mirrored in other model organisms. We are currently in the process of obtaining necessary human trial permissions, and should have our planned experiment initiated within 80 years, pending expedited review by local and Federal IRBs.

ACKNOWLEDGEMENTS:

I am grateful for generous support from the National Research Foundation, Black Hole Diet Plans, and the High Fructose Sugar Association. Transport flights were funded by SPACE-EXES, the consortium of wives divorced from insanely wealthy space-flight startups. I am also grateful for comments on early drafts by Mañana Athletic Club, Corpus Christi, USA. Finally, sincere thanks to the Cuy Foundation for generously donating animal care after the conclusion of the study.

LITERATURE CITED:

NASA. 1982. Project STS-XX: Guinea Pigs. Leaked internal memo.
Sekulić, S.R., D. D. Lukač, and N. M. Naumović. 2005. The Fetus Cannot Exercise Like An Astronaut: Gravity Loading Is Necessary For The Physiological Development During Second Half Of Pregnancy. Medical Hypotheses. 64:221-228
Xavier, M. 1965. Elastane Purchases Accelerate Weight Gain In Case-control Study. Journal of Obesity. 2:23-40.

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CONSTRUCTION & SIMULATION ANALYSIS OF AN IMPROVED ACTIN FILAMENT MODEL

Here, we present a new model of the actin filament (F-actin) that incorporates the global structure of a recently published model[1] but also conserves internal stereochemistry. The improved quality of the model is apparent in a comparison made of the model with other recent F-actin models using molecular dynamics (MD) simulation, monitoring a number of structural determinants.

In addition, simulations of the model are carried out in states with both ADP or ATP bound and local hydrogen-bonding differences characterized. The results point to the significance of a direct interaction of Glu137 with ATP for activation of ATPase activity after the G-to-F-actin transition.

Jeremy C. Smith

Thomas Splettstoßer

Kenneth C. Holmes

Frank Noé

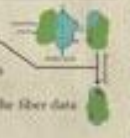
INTRODUCTION

The atomic-detail structure of F-actin is still unknown. We propose a new model of F-actin, "Holmes 2010", which was built using a straightforward approach in which priority was given to keeping the stereochemistry within the actin protomer intact while altering the position of the two actin domains to account for the global conformational change during the G-to-F-actin transition. A comparison is made of the structures and dynamics of Holmes 2010 with Oda 2009[2] and Holmes 2004[3] by subjecting them to MD simulation.

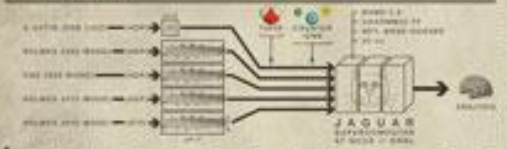
The nucleotide effects (ADP or ATP) on the conformation of the filament are also studied with a particular focus on the G-to-F-actin ATPase activation.

HOLMES 2010 MODEL

- Starting structure: G-actin X-ray structure PDB 1J6Z[4]
- Global conformational change during G→F-actin transition
- Separate fitting of two domains to Oda model
- DNase I binding loop: replace α-helix with Oda coordinates
- Phalloidin added and coordinates refined against fiber diffraction data[5] and EM data,[6] weighted in favor of the fiber data
- Final radius of gyration: 31.7 Å



METHODS



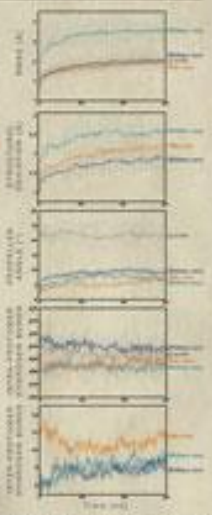
RESULTS

The new Holmes 2010[1] model was developed and MD simulations of 30 ns length were performed to compare the structure and stability with those of Oda 2009 and Holmes 2004. A further simulation of the Holmes 2010 model was performed in the ATP state.

Most properties were assessed on the protomer level and then averaged over the 13 protomers in the filament.

The average RMSD of the Oda 2009 and Holmes 2010 model protomers is very close to that of the simulation of the G-actin crystal structure. Among the 3 tested models, the structural deviation of Holmes 2004 is the lowest and that of Holmes 2010 the highest. As expected, the propeller angle formed by the four subunits of the actin molecule is much higher in the G-actin simulation than in the simulations of the F-actin models.

The number of hydrogen bonds within the actin molecules is the highest in the MD of Holmes 2010, even higher than that of G-actin. But most hydrogen bonds between protomers are formed in the MD of Oda.



Among the three models, only Holmes 2004 has a DNase binding loop in α-helical conformation, which unfolds entirely in 6 of the 13 protomers. Further, all protomers in Holmes 2004 show unfolding of some of the secondary structural elements. In the simulations, the hydrophobic plug showed a high degree of structural variation in Holmes 2004 and some variation in the Oda 2009 MD but was highly conserved among the Holmes 2010 protomers. Residues 348-357 form an α-helix that remains stable in the Oda simulations but unfolds in some protomers of the two Holmes simulations. The WH2 binding domain (residues 165-172) forms a β-hairpin in the simulations of both Holmes models but a bend in the Oda MD. Residues 227-237 exhibit partial unfolding in most Oda protomers.

Previous studies suggest the significance of Glu137 for ATP hydrolysis.[7,8] The twist of the actin monomer upon integration into the filament brings Glu137 close to ATP. In the MD Glu137 forms a stable H-bond with the oxygen of the β-phosphate group, and may thus play a direct role in hydrolysis.

CONCLUSIONS

The MD comparison reflects the evolution in quality of the actin models over the last years. The Holmes 2010 model possesses both a global conformation in agreement with the recent Oda model and a consistent intraprotomer stereochemistry. As such it should form a useful basis for atomic-detail investigations into F-actin structure-dynamics-function relationships.

REFERENCES

[illegible]

ACKNOWLEDGEMENTS

We thank Isabella Daidone and Durba Sengupta for helpful discussion and acknowledge support from Deutsche Forschungsgemeinschaft and a DOE Research and Development "Systems Biology" Grant. Computations were performed on the Jaguar Cray XT4 supercomputer at the National Center for Computational Sciences at Oak Ridge National Lab.

OAK RIDGE National Laboratory

Use Portrait Layout

MIRACINAE IN AUSTRALIA: Investigating the biodiversity of an unexplored subfamily of parasitoid wasps

Mollie Slater-Baker, Erinn Fagan-Jeffries
University of Adelaide

MIRACINAE

Miricinae is small subfamily of braconid wasps, characterised by their 14-segmented antennae, a distinctive 'Y' shaped structure on the abdomen, and reduced wing venation. They are parasitoids of the leaf-mining larvae of small moths, and are known to inject their host with symbiotic polydnaviruses when the female lays her eggs, acting to suppress the host immune system.

Actual size!

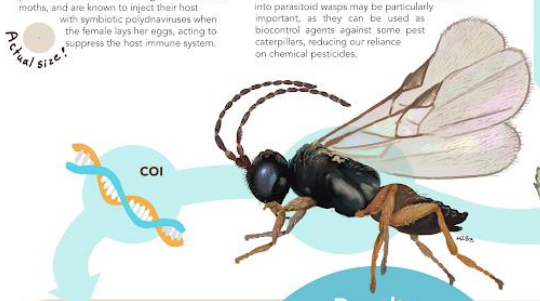
WHY STUDY THEM?

Research on Miracinae is currently very limited, and to date, no work has been done on these wasps in Australia. Describing and studying biodiversity is important for a variety of reasons, from conservation to human health. Research into parasitoid wasps may be particularly important, as they can be used as biocontrol agents against some pest caterpillars, reducing our reliance on chemical pesticides.

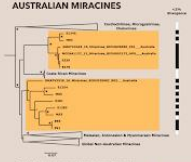
AIM: To perform a preliminary exploration of the biodiversity of Australian miracine wasps.

METHODS

DNA was extracted from Australian miracines, and the cytochrome c oxidase subunit I (COI) region was Sanger sequenced. Australian sequence data was combined with all Miracinae COI sequences from the Barcode of Life Database, to perform phylogenetic analyses and species delimitation.



PHYLOGENY OF AUSTRALIAN MIRACINES



Scan here for additional figures

Results

- Australian miracines form two distinct clades, most closely related to specimens from Costa Rica, the Malay archipelago and Myanmar.
- Australian miracines appear to be a basal lineage in the global context
- Of 14 Australian Specimens, 10 distinct species were recognised.

SAME SPECIES?

SPOT THE DIFFERENCE!

Scan here for specimen images

CONCLUSIONS

- The number of distinct specimens in a limited sample suggests that Australia may have many endemic species of Miracinae
- Morphological differences between delimited species were difficult to characterise, and some species may be morphologically cryptic
- DNA extraction was unsuccessful for older samples collected in 2005, so specimen age may be a limiting factor for DNA sequencing

WHAT NEXT?

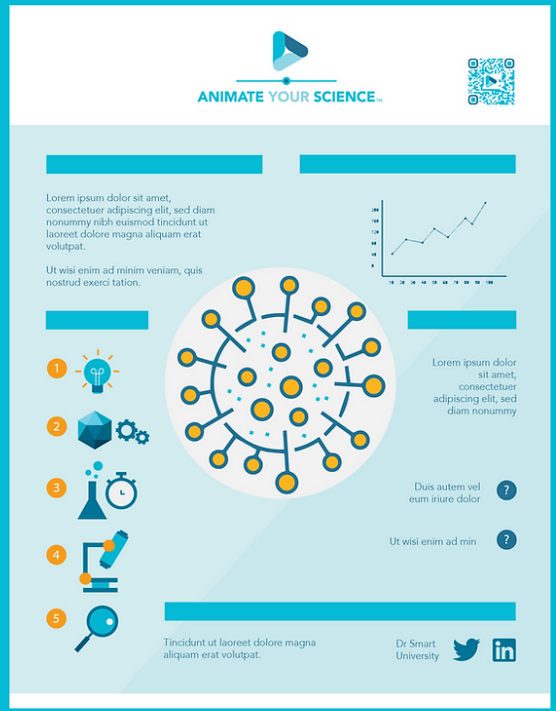
There is much more to be done towards understanding the Australian Miracinae, and further avenues of study may involve:

- Increased sampling and barcoding of miracines from across Australia
- Phylogenetic analysis using additional loci such as internal transcribed spacer 2 (ITS2) and wingless (WG) to produce a more robust phylogeny for the group
- Detailed imaging of morphological features using scanning electron microscopy
- Formal identification and description of Australian species

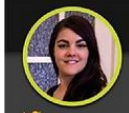
~~Text~~

I can barely read that!

Ooh, that's pretty!



Nice Color Scheme



SIZING UP THE CRAB INVADERS!

Morphometrics for *Carcinus* spp. can assess population variability & may assist with species identification

@reneecampbellart

René Campbell[1], Sabine Dittmann[1], Michael Gardner[1], Marty Deveney[2]

rene.campbell@flinders.edu.au

CRUSTACEAN DOMINATION

European shore crabs - some of the most widespread marine invasive species worldwide. Two species in the genus:

Carcinus maenas (Atlantic) & *C. aestuarii* (Mediterranean)

Prey on native & commercial species = impacts!

WHICH CRAB IS WHICH?

Taxonomy & population biology: may assist management
Morphometrics: quantitative analysis of shape & form

Carcinus species not confirmed in South Australia

Can assess taxonomic ID & local population variability

Populations: Variation due to life-history or plasticity

Species ID: Potential insight into hybridisation, adaptation or source populations

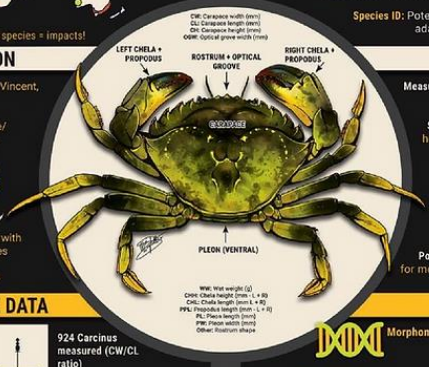
CRAB COLLECTION

Field sites: Coast of Gulf St Vincent, South Australia

Habitat regions: Rocky shore/ estuary, port & mangroves

Specimens: Crabs collected with baited traps & time searches

Seasonal from 2012 - 2018



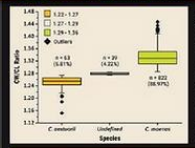
MEASUREMENTS

Measurements: Digital callipers used for morphometrics

Species ID: Carapace ratios can help identify species (based on previous literature)

Population: Multivariate analyses for morphometrics between regions & sex assessed

A LOOK AT SOME DATA

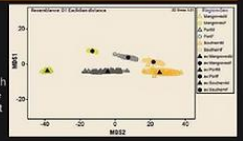


924 Carcinus measured (CW/CL ratio)

Higher proportions of *C. maenas* than *C. aestuarii* ($p < 0.05$)

Species not certain as female ratios are more variable

MDS + bootstrap averages for 12 morphometric features



Differences for both sex & region - male crabs most distinct in mangroves

CW & CL - strongest drivers behind these differences
Mangrove crabs tend to be larger = more distinct grouping in MDS

SIMPER Analysis

Mangroves mean similarity	Port mean similarity	Rocky shore mean similarity
88.85%	84.92%	79.69%

Major contributions - carapace width (~18%), carapace length (~14.5%), right chela length (~9.5%)

WHAT'S NEXT?

Morphometrics - influenced by sex, so species ID inconclusive

Population: Morphology distinct across regions + sex

Potential adaptations to habitats + life history (dispersal)

Population genetics will be assessed - connectivity & gene flow

Species ID: CW/CL ratios suggest both species are present in SA

Sex & intraspecific variability can influence ratios - unreliable?

DNA: species being confirmed with DNA barcoding (COI gene)

IMPLICATIONS

SA's Mediterranean climate - ideal for *C. aestuarii* establishment, or plasticity for *C. maenas*

Habitat use & potential range expansion - concerns for management of this global invader

[1]College of Science & Engineering, Flinders University, Sturt Road, Bedford Park, SA 5042
[2]South Australian Research & Development Institute, West Beach, SA 5024

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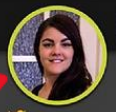
Make it Personal



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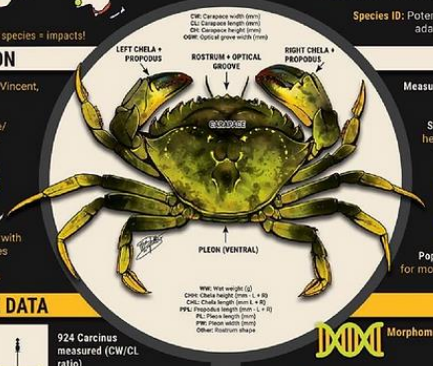

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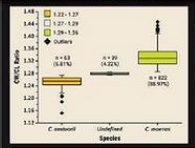

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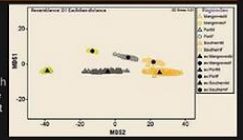

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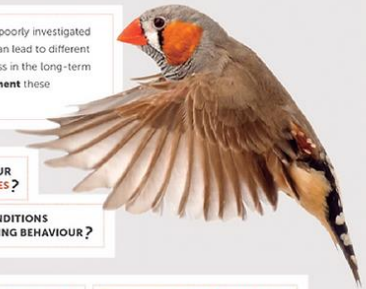
Eye catching main graphic

FLEXIBLE FORAGING BEHAVIOUR IN WILD ZEBRA FINCHES AND ITS RELATION WITH TEMPERATURE

CATERINA FUNGHI, LUKE MCCOWAN, WIEBKE SCHUETT AND SIMON GRIFFITH

INTRODUCTION

- **Foraging as behavioural trait** has been poorly investigated
- **Fluctuation** in enviromental contitions can lead to different behavioural strategies being equal in fitness in the long-term
- In **extreme and unpredictable environment** these fluctuations are more pronounced



QUESTIONS

IS INDIVIDUAL FORAGING BEHAVIOUR CONSISTENT IN WILD ZEBRA FINCHES?

HOW DO THE ENVIRONMENTAL CONDITIONS (TEMPERATURE) INFLUENCE FORAGING BEHAVIOUR?

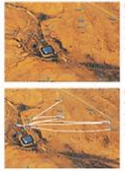
MATERIALS AND METHODS

16 FEEDERS RANDOMLY LOCATED (I.E. 2 TRIALS)

Zebra finches were PIT - tagged and foraging behaviour monitored using a decoder-antenna-feeder system over 3 weeks.

PC ANALYSIS ON DAILY FORAGING BEHAVIOUR OF 72 ADULTS

52% variation	PC1 foraging
Visits	0.3
Distance travelled	0.58
Nb. feeders	0.6
Feeder fidelity	-0.46



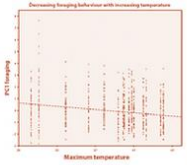
Example of 2 individuals showing different foraging behaviour. Dots represent feeders located 800m around a dam

LOW PC1 foraging:
- visits per feeder
- distance travelled
- feeders visited
- fidelity

HIGH PC1 foraging:
- visits per feeder
- distance travelled
- feeders visited
- fidelity

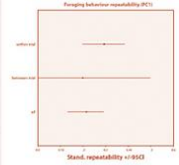
RESULTS AND DISCUSSION

IN EXTREME ENVIRONMENTS FORAGING BEHAVIOUR WAS INFLUENCED BY TEMPERATURE



GLMM: P=0.002, N=68. Negative relation between foraging behaviour (PC1) and Temperature (°C)

CHANGED ENVIRONMENT LED TO FLEXIBLE FORAGING BEHAVIOUR > OPPORTUNISM?



Repeatability (LMM) calculated considering all days of all trials, between and within trials.

Universität Hamburg
MACQUARIE University

Contact:
caterina.funghi@students.mq.edu.au

Put your Group ID & Names on the Poster